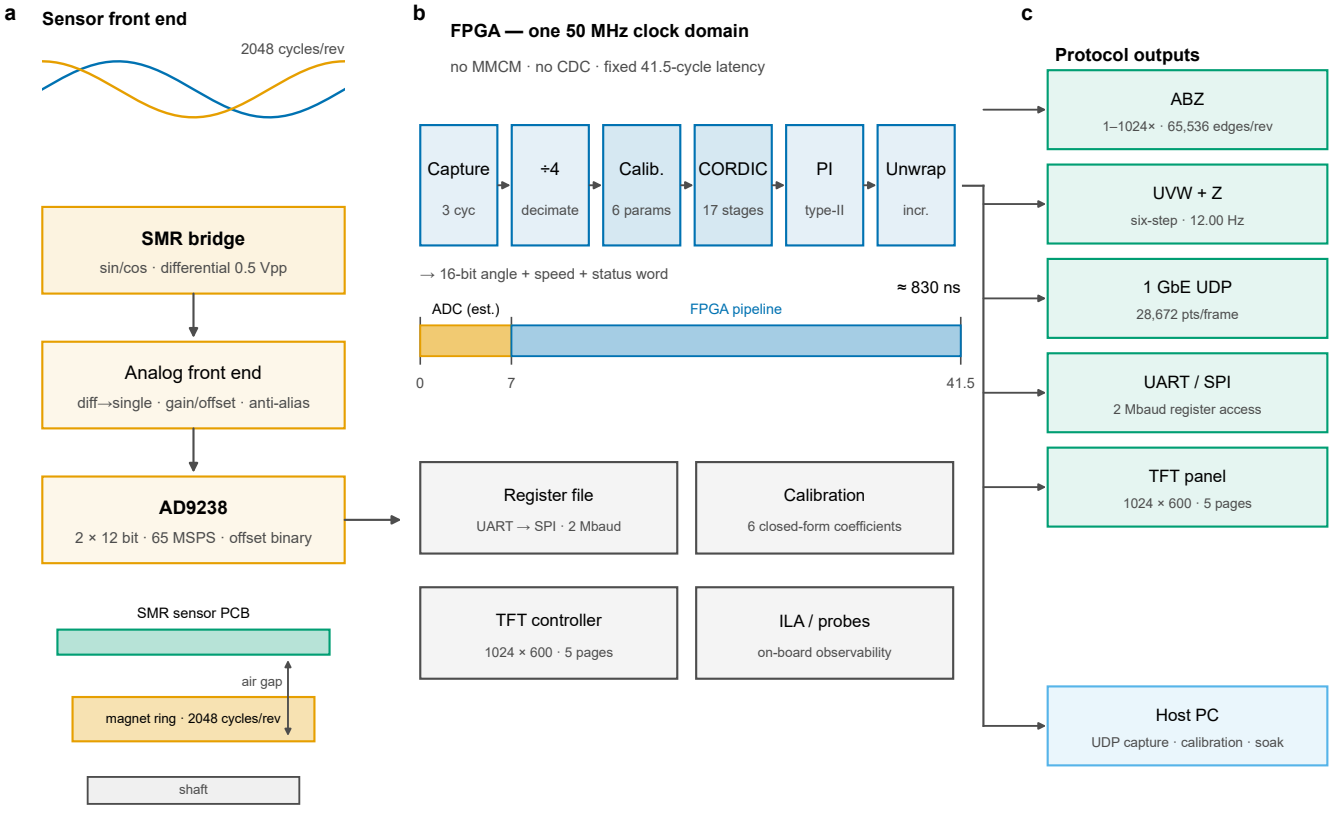

a
Sensor front end
2048 cycles/rev
SMR bridge
sin/cos · differential 0.5 Vpp
Analog front end
diff→single · gain/offset · anti-alias
AD9238
2 × 12 bit · 65 MSPS · offset binary
SMR sensor PCB
air gap
magnet ring · 2048 cycles/rev
shaft
b
FPGA — one 50 MHz clock domain
no MMCM · no CDC · fixed 41.5-cycle latency
Capture
3 cyc
÷4
decimate
Calib.
6 params
CORDIC
17 stages
PI
type-II
Unwrap
incr.
→ 16-bit angle + speed + status word
≈ 830 ns
ADC (est.)
FPGA pipeline
0
7
41.5
Register file
UART → SPI · 2 Mbaud
Calibration
6 closed-form coefficients
TFT controller
1024 × 600 · 5 pages
ILA / probes
on-board observability
c
Protocol outputs
ABZ
1–1024× · 65,536 edges/rev
UVW + Z
six-step · 12.00 Hz
1 GbE UDP
28,672 pts/frame
UART / SPI
2 Mbaud register access
TFT panel
1024 × 600 · 5 pages
Host PC
UDP capture · calibration · soak